Bayesian Transfer Learning for Object Detection in Optical Remote Sensing Images

Changsheng Zhou, Jiangshe Zhang, Junmin Liu, *Member, IEEE*, Chunxia Zhang, Guang Shi, and Junying Hu

***Abstract*— In the literature of object detection in optical remote sensing images, a popular pipeline is first modifying an off-the-shelf deep neural network, then initializing the modified network by pretrained weights on a source data set, and finally fine-tuning the network on a target data set. The procedure works well in practice but might not make full use of underlying knowledge implied by pretrained weights. In this article, we propose a novel method, referred to as Fisher regularization, for efficient knowledge transferring. Based on Bayes' theorem, the method stores underlying knowledge into a Fisher information matrix and fine-tunes parameters based on the knowledge. The proposed method would not introduce extra parameters and is less sensitive to hyperparameters than classical weight decay. Experiments on NWPUVHR-10 and DOTA data sets show that the proposed method is effective and works well with different object detectors.**

***Index Terms*— Bayes' theorem, deep learning, object detection, remote sensing, transfer learning.**

I. Introduction

RECENTLY, many object detection approaches based on deep learning have sprung up, thanks to the appearance of large-scale image data sets, such as ImageNet (IN) [1], and COCO [2]. Generally speaking, these methods could be organized into two groups: two-stage methods, which include an individual step to generate region proposals, and one-stage methods, which could be trained in an end-to-end manner. Typically, two-stage methods could get high accuracy but slow detection speed, for example, regions with convolutional neural network features (R-CNN) [3], Fast R-CNN [4], Faster R-CNN [5]. Conversely, one-stage methods could get almost real-time detection but low accuracy than two-stage methods, for example, YOLO [6], YOLOv2 [7], YOLOv3 [8].

This trend has been promoting object detection in optical remote sensing images (RSIs) heavily. For example, leveraging R-CNN together with hard-negative mining and deconvolutional operation, Tang *et al.* [9] proposed a novel method to detect vehicles in aerial images. Yang *et al.* [10] presented R^2CNN++ leveraging Faster R-CNN and attention mechanism. Applying one-stage method Single Shot multibox Detection (SSD) [11], Chen *et al.* [12] proposed recurrent detection with activated semantics (RDAS) by adding an auxiliary segmentation module. Tayara and Chong [13] and Yang *et al.* [14] leveraged an advanced technique, feature pyramid networks (FPNs) [15], to improve the detection accuracy further. By leveraging multiscale and rotation-insensitive features from convolutional layers, Wu *et al.* [16] proposed multi-scale and rotation-insensitive convolutional channel features (MsRi-CCF) which could yield robust detection results. Another impressive and important work in the literature is DOTA data set [17]. It is a large-scale expert-annotated object data set with 15 categories in the Earth vision. It is challenging for its tremendous object instances, arbitrary but well-distributed orientations, various categories, and complicated aerial scenes. Its appearance gives an opportunity for training very deep neural networks.

Despite these achievements, how to leverage knowledge learned from one data set (or task) is still in the air. A popular pipeline would first modify off-the-shelf networks, then use pretrained weights on IN data set to initialize networks, and finally fine-tune weights on a target data set [18]–[20]. It gets rid of pain from reinventing a whole network from scratch, but no one knows whether the procedure is reasonable or not. By assuming that trained models could be corrected adaptively from training data set to testing data set, Zou and Shi [21] presented random access memory (RAM) algorithm, which leverages a closed-form solution to fine-tune a learned detector when testing. Unlike other tasks in remote sensing, such as unmixing [22], [23], data fusion [24], [25], classification [26], [27], and object detection in optical RSIs usually only take RBG channels into consideration, which is consistent with the detection tasks in natural images for reusing off-the-shelf networks easily.

Based on Bayes' theorem, we propose a novel method to transfer knowledge from a source data set to a target data set for object detection in optical RSIs. Basically, our aim is to learn a new detector according to an existing detector and a target data set. To summarize, the main contributions of this article are highlighted as follows.

1) A one-stage method, you only look once version 3 (YOLOv3), is applied to object detection in optical RSIs, but with some modifications. Specifically, a

Manuscript received August 18, 2019; revised December 22, 2019 and February 5, 2020; accepted March 20, 2020. Date of publication April 14, 2020; date of current version October 27, 2020. This work was supported in part by the National Natural Science Foundation of China under Grant 61976174, Grant 61877049, Grant 11671317, and Grant 11991023 and in part by the National Key Research and Development Program of China under Grant 2018AAA0102201. *(Corresponding authors: Jiangshe Zhang; Junmin Liu.)*

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Digital Object Identifier 10.1109/TGRS.2020.2983201

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class-balanced loss function is introduced due to the imbalanced amount of objects of different categories, mean-squared error loss is substituted by smooth L_1 loss for training stability in early training stage, and four different anchors is assigned in each output head for handling a wide variety of object size in RSIs.

2) A novel Fisher regularization algorithm is proposed. The method stores knowledge from a source data set into a Fisher Information Matrix (FIM) and its trivial solution is exactly corresponding to the "pretraining initialization." Moreover, it would not introduce extra parameters, and it is less sensitive to hyperparameters than the classical method weight decay.
3) A useful property is found to accelerate the procedure of matching ground-truth bounding boxes with predicted ones. For a given image, the number of calculation of intersection over union (IoU) for matching could decrease from ten thousands of times to hundreds of or even tens of times. It is time-saving during training.
4) Experiments on NWPUVHR-10 and DOTA data sets show that the proposed method improves the performance of one-stage detectors significantly and goes more closed to two-stage detectors. Moreover, it works well with different object detectors.

The rest of this article is organized as follows. In Section II, related works are discussed. In Section III, the Bayesian framework of object detection and the proposed method are stated particularly. In Section IV, experiments of the proposed method are given together with comparison among different methods. Finally, this article is concluded in Section V.

II. Background

In this section, we discuss some previous works mostly related to this article, covering the network architecture to be used and the inspiration of our method. These works include YOLOv3, RAM, and Bayes' theorem.

A. YOLOv3

YOLOv3 [8] is a representative work of one-stage methods for object detection. On the one hand, it takes many good ideas from other works, such as skip connection in ResNet [28], feature pyramid in FPN [15], and multiscale training [7]. On the other hand, it develops some novel and useful ideas, such as assigning only one predicted box for each ground-truth box, using binary cross entropy loss (BCELoss) for classification, and using k-means clustering to determine prior anchors according to data sets.

Fig. 1 shows the network architecture of YOLOv3. It is a kind of fully convolutional networks [29]. Basically, it could be divided into three parts: backbone part, body part, and head part. Usually, the backbone part is a deep network for classification and pretrained on IN. The body part is a FPN-style [15] architecture for multiscale feature fusion. The head part has three outputs, with the top one for large objects, the middle one for medium objects, and the bottom one for small objects.

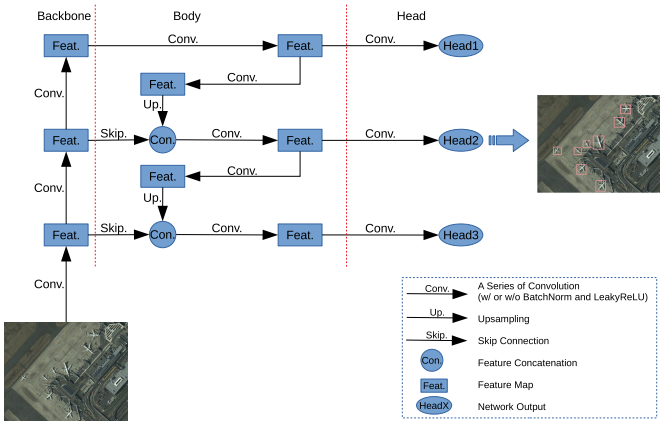


Fig. 1. Network architecture of YOLOv3. Except for the head part, we do not modify the architecture any further. See the main body of the text for more details.

When predicting, YOLOv3 outputs four offsets according to a prior anchor. Mathematically, for network output values $(\hat{x}_t, \hat{y}_t, \hat{w}_t, \hat{h}_t)$ and a prior anchor (x_a, y_a, w_a, h_a), the corresponding predicted box could be expressed as

$$\hat{x} = \sigma(\hat{x}_t) + x_a, \quad \hat{w} = w_a e^{\hat{w}_t}$$
$$\hat{y} = \sigma(\hat{y}_t) + y_a, \quad \hat{h} = h_a e^{\hat{h}_t}$$

where $(\hat{x}, \hat{y})$ stands for the center of the predicted box, and $(\hat{w}, \hat{h})$ for the width and height of the predicted box, respectively.

Some modifications are made for detecting objects in optical RSIs. Specifically, multiscale training is replaced by SSD-style data augmentation [11] for time-saving consideration. Four anchors are used in every output head to deal with the wide variety of object size. Smooth L_1 Loss (SmoothL1Loss) [4] is used for training stability in early stage. Class-balanced loss is leveraged due to the imbalanced amount of objects of different categories.

B. RAM

RAM is an algorithm for dynamically updating learned models according to test samples [21]. This method breaks traditional paradigm of "first training a model and then fixing it for testing." It does not need iterative update. When its loss function is SmoothL1Loss, RAM aims at updating trained model by an approximated closed-form solution

$$\mathbf{w}_{\text{new}} = (\mathbf{C} + \alpha\mathcal{I})^{-1}\left(E_{(\mathbf{x},y)\sim\mathcal{D}_{\text{tr}}}[y\mathbf{x}] + \alpha\mathcal{I}\hat{\mathbf{w}}\right)$$

where $\mathbf{w}_{\text{new}}$ is the updated parameters, α is a hyperparameter, $(\mathbf{x}, y)$ stands for a pair of data point and the corresponding label, $\mathcal{D}_{\text{tr}}$ is training data set, and $\hat{\mathbf{w}}$ is the parameters of the trained model. Moreover, $\mathcal{I}$ and $\mathbf{C}$ are computed as follows:

$$\mathcal{I} = E_{(\mathbf{x},y)\sim\mathcal{D}_{\text{tr}}}\left[\zeta(\mathbf{x}, y)\mathbf{x}\mathbf{x}^T\right]$$
$$\mathbf{C} = E_{(\mathbf{x},y)\sim\mathcal{D}_{\text{ob}}}\left[\zeta(\mathbf{x}, y)\mathbf{x}\mathbf{x}^T\right]$$
$$\zeta(\mathbf{x}, y) = \begin{cases} 1, & |\hat{\mathbf{w}}^T\mathbf{x} - y| < 0.1 \\ 0, & \text{otherwise} \end{cases}$$

where $\mathcal{D}_{\text{ob}}$ stands for test set.

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RAM is not very suitable for knowledge transferring. First, it requires that training set and test set are of the same type, that is, samples should come from an individual distribution. Second, it neglects positive latent variables (objects of interest) for constraints relaxing. Last, the update rule of RAM is an approximation solution and would be invalid when objective function changes.

Different from RAM, transferring knowledge between different kinds of data sets is our standpoint. Based on the knowledge learned from source data set, we derive a regularization term, which would take part in gradient calculation to guide the learning procedure on target data set.

C. Bayes' Theorem

Formally, Bayes' theorem offers a way to reverse conditional probability, and it could be expressed as follows:

$$p(M|D) = \frac{p(D|M)p(M)}{p(D)}$$

where usually M is an unobservable latent variable, and D is an observable variable. On the left-hand side, $p(M|D)$ is called posterior probability. On the right-hand side, $p(D|M)$, $p(M)$, and $p(D)$ are usually called likelihood, prior, and evidence, respectively. Because the denominator only scales the posteriori to make it a proper probability, Bayes' theorem usually is simplified as

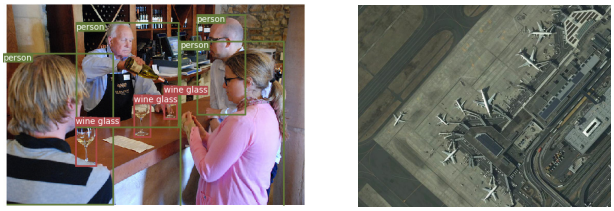
$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

where $\propto$ stands for "is proportional to".

Bayes' theorem has achieved great success in machine learning. From naive Bayes for spam emails classification [30] to hidden Markov model for speech recognition [31], and from conditional random field for image labeling [32] to Bayesian networks for gene expression analyzing [33], Bayes' theorem has been applied to various fields in practice.

Many works in deep learning also take advantage of Bayes' theorem. By integrating Bayesian models and deep learning together, Wang and Yeung [34] proposed a principled probabilistic framework, called Bayesian deep learning. Based on past experiences, Wang *et al.* [35] proposed a new framework to estimate Gaussian process prior for meta Bayesian optimization. By leveraging a Markov random field-based prior, Zhang *et al.* [36] proposed a model spectrum correlation and pixel similarity together for hyperspectral image reconstruction. Moreover, Bayes' theorem has also been applied to 1-bit convolutional neural networks optimization [37] and neural architecture search [38]. These works show that Bayes' theorem could still play a crucial role in the literature of deep learning.

Different from above, Bayes' theorem is introduced to model the procedure of knowledge transferring from source data set to target data set in this article. It helps to derive a "loss + regularization" style objective function and gives a proper interpretation about "pretraining initialization." More details would be given in the following section.



(a) (b)

Fig. 2. Detection results based on a detector trained on COCO data set. (a) Test sample from COCO. (b) Test sample from NWPUVHR-10.

III. Methodology

Formally, the learning procedure of an object detector is a procedure of solving a maximum likelihood estimation (MLE) problem. Given a source data set D_S with images I_S and the corresponding bounding boxes B_S for training, an object detector $\hat{f}_S$ could be determined by solving the following MLE problem

$$\hat{f}_S = \arg\max_{f \in \mathcal{F}} p(D_S|f) \tag{1}$$

where $\mathcal{F}$ stands for a set of detection models.

Usually, the learned detector is directly used to predict bounding boxes $\hat{B}_S$ of a new data point I_{new}:

$$\hat{B}_S = \hat{f}_S(I_{new}).$$

This prediction procedure works well when I_{new} is similar to samples in the source data set D_S.

However, when I_{new} differs from samples in the source data set, the prediction procedure often fails. In this case, for example, I_{new} may contain objects with new categories or with different scale, or may have varied background, or may be taken from a different camera view. Any of these variations could worsen the prediction. Fig. 2 illustrates the issue clearly. A detector trained on COCO data set [2] performs well in a similar natural image, but it completely fails to detect any object of interest in an optical RSI, which is from NWPUVHR-10 data set [18], [39]. This phenomenon is called "data bias" in this article.

There are two straightforward approaches to deal with data bias. One is to train a new detector from scratch based on a properly collected data set, but it could take a lot of time both for annotating and training. Another one is to initialize neural networks using pretrained weights. The pretraining initialization method is very popular and often achieves good results, but no one knows whether it could make full use of knowledge learned from source data set. Moreover, it could still perform terribly when training data are not sufficient, which is often the case for optical RSIs.

In this article, we propose a novel method for object detection in optical RSIs, called Bayesian transfer learning, where it is assumed that knowledge learned from source data set could benefit tasks on target data set. The method unifies pretraining initialization and fine-tuning into a Bayesian framework.

Given a detector $\hat{f}_S$ trained on a source data set $D_S = (I_S, B_S)$, and a target data set $D_T = (I_T, B_T)$ which is different from D_S, our goal is to learn a new detector $\hat{f}_T$ based on $\hat{f}_S$ and D_T. Here, S, T, I, B stand for "source, target,

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image, bounding box," respectively. Mathematically, $\hat{f}_{\mathrm{T}}$ could be obtained by solving the following maximum a posteriori estimation problem

$$\hat{f}_{\mathrm{T}} = \arg\max_{f\in\mathcal{F}} p\left(f|D_{\mathrm{T}}, \hat{f}_{\mathrm{S}}\right) \tag{2}$$

where $\mathcal{F}$ again denotes a set of detection models for simplicity and then the Bayes' theorem implies that

$$\begin{aligned} p\left(f|D_{\mathrm{T}}, \hat{f}_{\mathrm{S}}\right) &= \frac{p\left(f, D_{\mathrm{T}}|\hat{f}_{\mathrm{S}}\right)}{p\left(D_{\mathrm{T}}|\hat{f}_{\mathrm{S}}\right)} \\ &= \frac{p\left(f|\hat{f}_{\mathrm{S}}\right)p\left(D_{\mathrm{T}}|f, \hat{f}_{\mathrm{S}}\right)}{p(D_{\mathrm{T}})} \\ &= \frac{p\left(f|\hat{f}_{\mathrm{S}}\right)p(D_{\mathrm{T}}|f)}{p(D_{\mathrm{T}})} \\ &\propto p\left(f|\hat{f}_{\mathrm{S}}\right)p(D_{\mathrm{T}}|f). \end{aligned} \tag{3}$$

In the equalities above, it is assumed that target data set and detector learned from source data set have no effect on each other, and in other words, D_{T} and $\hat{f}_{\mathrm{S}}$ are independent. In the last row of (3), the first term $p(f|\hat{f}_{\mathrm{S}})$ is a model prior related to the pretrained model on source data set and the second term is the likelihood of target data set. Next, we will analyze these two terms.

A. Model Prior

As (3) shows, the model prior term $p(f|\hat{f}_{\mathrm{S}})$ need be determined first. Usually, there are various ways to choose a prior probability, such as Gaussian distribution for ridge regression, sample proportion for kernel density classification, or uniform distribution for model selection [40].

For simplicity, a Gaussian distribution is taken as the model prior, $\mathcal{N}(\mu, \Sigma)$, where μ, Σ are mean and covariance matrix, respectively. The question is how to determine μ and Σ. As simple as possible, it could be assigned that μ equals zero (vector) and Σ equals identity matrix. In this case, a common regularization term, weight decay, could be derived by taking logarithm operation on the model prior term, but it does not leverage $\hat{f}_{\mathrm{S}}$.

Another way is to choose $p(f|D_{\mathrm{S}})$ as the model prior directly, but the real probability is unknown. Based on the widely used framework Laplace approximation [41], $\log p(f|D_{\mathrm{S}})$ could be expanded as a Taylor series, when it has at least second-order derivative at $f = \hat{f}_{\mathrm{S}}$:

$$\begin{aligned} \log p(f|D_{\mathrm{S}}) &\approx \log p\left(\hat{f}_{\mathrm{S}}|D_{\mathrm{S}}\right) \\ &\quad + \left(f - \hat{f}_{\mathrm{S}}\right)^{\mathrm{T}} \nabla \log p(f|D_{\mathrm{S}})|_{f=\hat{f}_{\mathrm{S}}} \\ &\quad + \frac{1}{2}\left(f - \hat{f}_{\mathrm{S}}\right)^{\mathrm{T}} \nabla^2 \log p(f|D_{\mathrm{S}})|_{f=\hat{f}_{\mathrm{S}}} \left(f - \hat{f}_{\mathrm{S}}\right) \\ &= \left(f - \hat{f}_{\mathrm{S}}\right)^{\mathrm{T}} \nabla \log p(f|D_{\mathrm{S}})|_{f=\hat{f}_{\mathrm{S}}} \\ &\quad + \frac{1}{2}\left(f - \hat{f}_{\mathrm{S}}\right)^{T} \nabla^2 \log p(f|D_{\mathrm{S}})|_{f=\hat{f}_{\mathrm{S}}} \left(f - \hat{f}_{\mathrm{S}}\right) \\ &\quad + \text{constant}. \end{aligned} \tag{4}$$

To simplify the right-hand side of the above equation, a property about posteriori and likelihood should be introduced first.

Property 1: D_{S} is a source data set, and $\hat{f}_{\mathrm{S}}$ is an MLE solution as shown in (1). Given that the prior $p(f)$ is uniform and all related probabilities have at least seconder-order derivative with respect to f, then the following equalities hold:

$$\begin{aligned} \nabla \log p(f|D_{\mathrm{S}})|_{f=\hat{f}_{\mathrm{S}}} &= 0 \\ \nabla^2 \log p(f|D_{\mathrm{S}})|_{f=\hat{f}_{\mathrm{S}}} &= \nabla^2 \log p(D_{\mathrm{S}}|f)|_{f=\hat{f}_{\mathrm{S}}}. \end{aligned}$$

Proof: Based on the chain rule, we know that

$$p(D_{\mathrm{S}}, f) = p(f|D_{\mathrm{S}})p(D_{\mathrm{S}}) = p(D_{\mathrm{S}}|f)p(f).$$

To prove the first equality, we need to take first logarithm and then derivation at $f = \hat{f}_{\mathrm{S}}$:

$$\begin{aligned} &\nabla \log p(D_{\mathrm{S}}, f)|_{f=\hat{f}_{\mathrm{S}}} \\ &\quad = \nabla \log p(f|D_{\mathrm{S}})|_{f=\hat{f}_{\mathrm{S}}} + \nabla \log p(D_{\mathrm{S}})|_{f=\hat{f}_{\mathrm{S}}} \\ &\quad = \nabla \log p(D_{\mathrm{S}}|f)|_{f=\hat{f}_{\mathrm{S}}} + \nabla \log p(f)|_{f=\hat{f}_{\mathrm{S}}}. \end{aligned}$$

Here, $p(f)$ is uniform implies that $\nabla \log p(f)|_{f=\hat{f}_{\mathrm{S}}} = 0$, and $p(D_{\mathrm{S}})$ does not contain f implies that $\nabla \log p(D_{\mathrm{S}})|_{f=\hat{f}_{\mathrm{S}}} = 0$, so that

$$\nabla \log p(f|D_{\mathrm{S}})|_{f=\hat{f}_{\mathrm{S}}} = \nabla \log p(D_{\mathrm{S}}|f)|_{f=\hat{f}_{\mathrm{S}}}.$$

And "$\hat{f}_{\mathrm{S}}$ is an MLE solution" implies that

$$\nabla p(D_{\mathrm{S}}|f)|_{f=\hat{f}_{\mathrm{S}}} = 0 \Rightarrow \nabla \log p(D_{\mathrm{S}}|f)|_{f=\hat{f}_{\mathrm{S}}} = 0$$

so that the first equality holds.

To prove the second equality, we need to take the second-order derivation at $f = \hat{f}_{\mathrm{S}}$:

$$\begin{aligned} &\nabla^2 \log p(D_{\mathrm{S}}, f)|_{f=\hat{f}_{\mathrm{S}}} \\ &\quad = \nabla^2 \log p(f|D_{\mathrm{S}})|_{f=\hat{f}_{\mathrm{S}}} + \nabla^2 \log p(D_{\mathrm{S}})|_{f=\hat{f}_{\mathrm{S}}} \\ &\quad = \nabla^2 \log p(D_{\mathrm{S}}|f)|_{f=\hat{f}_{\mathrm{S}}} + \nabla^2 \log p(f)|_{f=\hat{f}_{\mathrm{S}}}. \end{aligned}$$

Similarly, we have $\nabla^2 \log p(D_{\mathrm{S}})|_{f=\hat{f}_{\mathrm{S}}} = 0$ and $\nabla^2 \log p(f)|_{f=\hat{f}_{\mathrm{S}}} = 0$, so that the second equality holds. ■

Property 1 implies that with those assumptions, (4) could be expressed as follows:

$$\begin{aligned} &\log p(f|D_{\mathrm{S}}) \\ &\quad \approx \frac{1}{2}\left(f - \hat{f}_{\mathrm{S}}\right)^{T} \nabla^2 \log p(D_{\mathrm{S}}|f)|_{f=\hat{f}_{\mathrm{S}}} \left(f - \hat{f}_{\mathrm{S}}\right) + \text{constant}. \end{aligned} \tag{5}$$

However, it is still hard to compute the second-order derivative of log-likelihood. Fortunately, the derivative could be substituted by FIM under some regularity conditions, as shown below.

Property 2: D_{S} is a source data set, $\log p(D_{\mathrm{S}}|f)$ is the corresponding log-likelihood function, and assume that all related probabilities have at least second-order derivative with respect to f. For any $z \in D_{\mathrm{S}}$, some definitions are as follows.

1) Score function, $S(f) = \nabla \log p(z|f)$.
2) FIM, $\mathcal{I}(f) = \mathrm{E}[S(f)S(f)^{\mathrm{T}}]$, or simply denoted as $\mathcal{I}$.
3) Hessian matrix, $H(f) = \nabla S(f) = \nabla^2 \log p(z|f)$.

Under some regularity conditions [42], the expectation of Hessian matrix equals the negative FIM:

$$\mathrm{E}[H(f)] = -\mathcal{I}(f).$$

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Proof: By expanding the second-order derivative, we get

$$\begin{aligned} H(f) &= \nabla^2 \log p(z|f) = \nabla\left(p^{-1}(z|f)\nabla p(z|f)\right) \\ &= -p^{-2}(z|f)\nabla p(z|f)\nabla p(z|f)^T + p^{-1}(z|f)\nabla^2 p(z|f) \\ &= -S(f)S(f)^{\mathrm{T}} + p^{-1}(z|f)\nabla^2 p(z|f) \end{aligned}$$

so that

$$\begin{aligned} \mathrm{E}[H(f)] &= \int p(z|f)H(f)dz \\ &= \int p(z|f)(-S(f)S(f)^T + p^{-1}(z|f)\nabla^2 p(z|f))dz \\ &= -\mathrm{E}\left[S(f)S(f)^T\right] + \int \nabla^2 p(z|f)dz \\ &= -\mathcal{I}(f) + \nabla^2 \int p(z|f)dz \\ &= -\mathcal{I}(f). \end{aligned}$$

■

Property 2 shows that the expectation of Hessian matrix equals the expectation of covariance matrix of score function. For it is difficult to compute expectation, we use the empirical ones as alternatives

$$\begin{aligned} &\frac{1}{N}\sum_{i=1}^{N} \nabla^2 \log p(z_i|f) \\ &\quad \approx \mathrm{E}[H(f)] = -\mathcal{I}(f) \\ &\quad \approx -\frac{1}{N}\sum_{i=1}^{N} \nabla \log p(z_i|f)\nabla \log p(z_i|f)^T \end{aligned} \tag{6}$$

where $z_i = (I_{\mathrm{S}}^{(i)}, B_{\mathrm{S}}^{(i)}) \in D_{\mathrm{S}}$ and N is the number of data points.

Furthermore, it is usually assumed that data points are i.i.d. and that means

$$\begin{aligned} \nabla^2 \log p(D_{\mathrm{S}}|f) &= \nabla^2 \log \prod_{i=1}^{N} p(z_i|f) \\ &= \sum_{i=1}^{N} \nabla^2 \log p(z_i|f). \end{aligned} \tag{7}$$

According to (6) and (7), we could rewrite (5) as follows:

$$\log p(f|D_{\mathrm{S}}) \approx -\frac{1}{2}\left(f - \hat{f}_{\mathrm{S}}\right)^T N\mathcal{I}\left(\hat{f}_{\mathrm{S}}\right)\left(f - \hat{f}_{\mathrm{S}}\right) + \text{constant}$$

and after taking the exponential operation, we could get

$$p(f|D_{\mathrm{S}}) \approx \mathcal{N}\left(\hat{f}_{\mathrm{S}}, \left(N\mathcal{I}\left(\hat{f}_{\mathrm{S}}\right)\right)^{-1}\right). \tag{8}$$

It should be noted that the approximation requires that related probabilities must have at least second-order derivative at $f = \hat{f}_{\mathrm{S}}$. But, it usually does not meet the requirement when these probabilities are modeled by neural networks with Rectified Linear Unit (ReLU)-style activation function [43], [44]. However, we still argue that the approximation could be done without any hesitation, for $\mathcal{I}(\hat{f}_{\mathrm{S}})$ is positive semidefinite so that the approximation (8) does make sense [it would degrade into a uniform probability if $\mathcal{I}(\hat{f}_{\mathrm{S}})$ equaled zero (matrix)]. Moreover, the approximation works well in practice and it has also been adopted in deep learning [45]–[47].

Therefore, the model prior is finally chosen as

$$p\left(f|\hat{f}_{\mathrm{S}}\right) := \mathcal{N}\left(\hat{f}_{\mathrm{S}}, \left(N\mathcal{I}\left(\hat{f}_{\mathrm{S}}\right)\right)^{-1}\right).$$

Based on the prior, the proposed regularization term of Bayesian transfer learning could be derived as follows:

$$l_{\mathrm{reg}} = \alpha\left(f - \hat{f}_{\mathrm{S}}\right)^T \mathcal{I}\left(\hat{f}_{\mathrm{S}}\right)\left(f - \hat{f}_{\mathrm{S}}\right) \tag{9}$$

where unrelated constant has been ignored and N is absorbed into α as a nonnegative hyperparameter.

From the regularization term (9), it is easy to find that $\hat{f}_{\mathrm{S}}$ is a trivial solution of $l_{\mathrm{reg}} = 0$. When a neural network is chosen as object detector, $\hat{f}_{\mathrm{S}}$ could be represented by its weights. From this point of view, initialization by pretrained weights is better than randomness.

B. Data Likelihood

The second term in the last row of (3) stands for the likelihood of the target data set, that is, the loss term on the target data set. In this part, the subscript S or T would be ignored when it does not cause any confusion.

To make it clear, some notation are introduced in this article. An image is denoted by I, a bounding box is denoted by $B =$ (loc, obj, cls), among which location (loc) contains centroids (x, y), width w, and height h of B, that is, loc $= (x, y, w, h)$. Objectness (obj) is denoted by obj $= 1$ if B contains one object of interest; otherwise, obj $= 0$. Class (cls) stands for the category of the object (if exists) in B. Then the probabilistic model of object detection could be expressed as

$$\begin{aligned} &p(I, B|f) \\ &\quad = p(I|f)p(B|I, f) \propto p(B|I, f) \\ &\quad = p(\mathrm{cls}|\mathrm{obj} = 1, \mathrm{loc}, I, f)p(\mathrm{obj} = 1|\mathrm{loc}, I, f)p(\mathrm{loc}|I, f). \end{aligned} \tag{10}$$

Accordingly, loss term should be divided into three parts: classification loss, objectness loss, and location loss.

The product of the first two terms in the right-hand side of (10) reflects how probable that B contains an object of interest with a specific category, and hereinafter the product is called confidence of B.

1) Classification Loss: A bounding box with an object of interest would contribute to the classification loss. Taking class imbalance into consideration, we propose to use class-balanced loss. Formally, given an image I and a class $c (c = 1, 2, \ldots, C)$, the corresponding boxes are denoted by $\{B_i(c, I)|i = 1, \ldots, b(c, I)\}$, where $b(c, I)$ (or b for simplicity) is the number of boxes with category being c in image I. Conveniently, $b > 0$ (if it were not, the corresponding loss would be 0; so as the following). Then the corresponding class-balanced loss could be expressed as

$$l_{\mathrm{cls}}(c, I) = \frac{1}{b(c, I)}\sum_{i=1}^{b(c,I)} L_{\mathrm{cls}}\left(\hat{B}_i(c, I), B_i(c, I)\right)$$

where $\hat{B}_i(c, I)$ is the predicted counterpart of $B_i(c, I)$ and L_{cls} is a classification loss function, such as BCELoss. Then the

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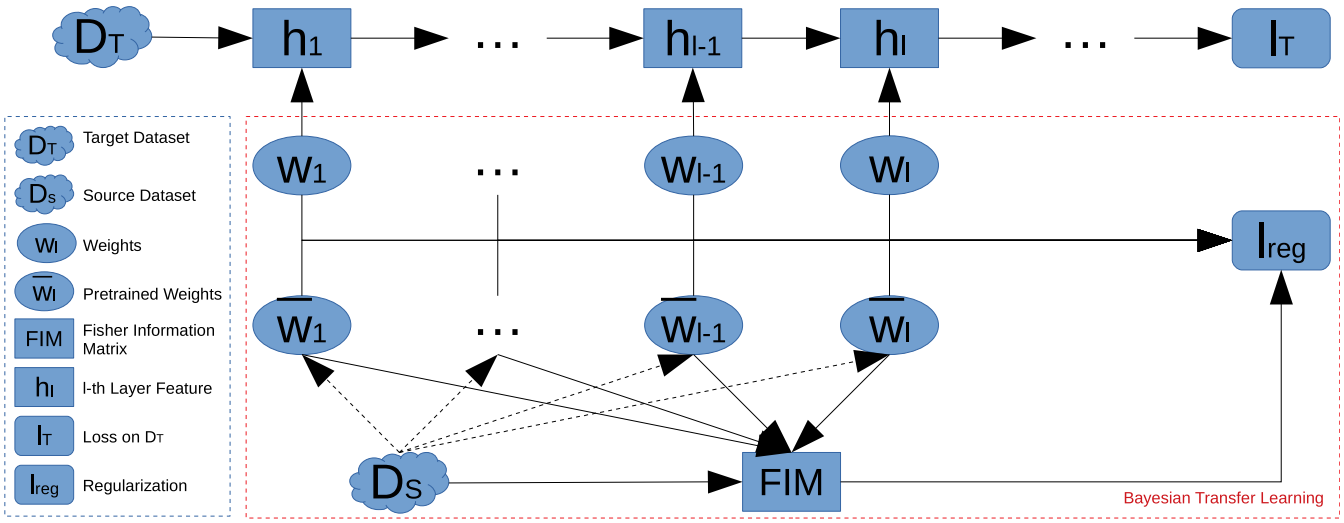


Fig. 3. Illustration of the proposed method. It contains two parts: one is the top line which illustrates traditional loss computation on target data set and another is the red box with dashed line which illustrates the proposed Bayesian transfer learning.

average classification loss of the image is

$$l_{\text{cls}}(I) = \frac{1}{C}\sum_{c=1}^{C} l_{\text{cls}}(c, I).$$

2) Objectness Loss: To calculate the objectness loss of a given image I, all predicted bounding boxes need be divided into three parts first: positive boxes, negative boxes, and ignored boxes. A positive box contains one and only one object of interest, while a negative ones does not. Ignored boxes satisfy some specific rules and do not contribute to any loss. Specifically, for a given ground-truth bounding box, we first compute its IoU with all predicted boxes. The box with maximum IoU is assigned to be positive; other boxes with IoU more than a threshold (e.g., 0.5 in this article) are ignored; the rest are negative [8]. For an image I, the number of total predictions is denoted by $b(I)$

$$b(I) = b(\text{pos}, I) + b(\text{neg}, I) + b(\text{ign}, I)$$

where $b(\text{pos}, I)$, $b(\text{neg}, I)$, and $b(\text{ign}, I)$ are the numbers of positive, negative, and ignored predictions, respectively. Then the objectness loss could be expressed as

$$l_{\text{obj}}(I) = \frac{1}{b(\text{pos}, I)}\sum_{i=1}^{b(\text{pos},I)} L_{\text{obj}}\big(\hat{B}_i(\text{pos}, I), B_i(\text{pos}, I)\big) + \frac{1}{b(\text{neg}, I)}\sum_{i=1}^{b(\text{neg},I)} L_{\text{obj}}\big(\hat{B}_i(\text{neg}, I), B_i(\text{neg}, I)\big)$$

where L_{obj} is a loss function for classification, such as BCELoss. It should be noticed that ignored boxes do not contribute to any loss.

3) Location Loss: Only positive boxes contribute to location loss. Mathematically

$$l_{\text{loc}}(I) = \frac{1}{b(\text{pos}, I)}\sum_{i=1}^{b(\text{pos},I)} L_x\big(\hat{B}_i(\text{pos}, I), B_i(\text{pos}, I)\big) + \frac{1}{b(\text{pos}, I)}\sum_{i=1}^{b(\text{pos},I)} L_y\big(\hat{B}_i(\text{pos}, I), B_i(\text{pos}, I)\big) + \frac{1}{b(\text{pos}, I)}\sum_{i=1}^{b(\text{pos},I)} L_w\big(\hat{B}_i(\text{pos}, I), B_i(\text{pos}, I)\big) + \frac{1}{b(\text{pos}, I)}\sum_{i=1}^{b(\text{pos},I)} L_h\big(\hat{B}_i(\text{pos}, I), B_i(\text{pos}, I)\big)$$

where L_x, L_y are mean square error loss (MSELoss) and L_w and L_h are SmoothL1Loss [4] in this article.

Then, the loss term on target data set could be expressed as

$$l_{\text{T}} = \text{E}\big[l_{\text{cls}}(I_{\text{T}}) + l_{\text{obj}}(I_{\text{T}}) + l_{\text{loc}}(I_{\text{T}})\big]. \tag{11}$$

C. Implementation Details

1) Algorithm: According to (2), (9), and (11), the final optimization problem could be formulated as

$$\hat{f}_{\text{T}} = \arg\min_f l_{\text{T}} + \alpha l_{\text{reg}} \tag{12}$$

where α is a nonnegative hyperparameter. Fig. 3 shows the computation procedure of our proposed method, in which Bayesian transfer learning plays a crucial role. It is called Fisher regularization algorithm (FisherReg) because FIM appears in the regularization term. Algorithm 1 gives more details about training procedure.

2) Fisher Information Matrix: The number of entries of FIM is proportional to the number of learnable parameters. FIM would be too big to be calculated or stored efficiently when there are millions of parameters, which is usually the case in the literature of deep learning. To prevent the issue, we use two simple but effective alternatives in experiments: one is called FIM-keynode, in which only some selected parameters are taken into consideration, and another one is called FIM-diagonal, in which only the diagonal entries of FIM are taken into consideration. In the case of FIM-keynode,

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Algorithm 1 Training Procedure of Fisher Regularization Algorithm

Require: Pretrained weights $\hat{w}$ and FIM $\mathcal{I}$ on source data set, batch size bs, learning rate η, and hyperparameter α. Training pairs of images and labels, (I, B).

1: **while** Objective function (12) has not converged **do**
2: Sampling a batch of training pairs, $\{(I_i, B_i)\}_{i=1}^{bs}$
3: $g_w^l \leftarrow \nabla_w \frac{1}{bs} \sum_{i=1}^{bs} l_T(I_i, B_i)$
4: $g_w^{reg} \leftarrow \nabla_w l_{reg}(\hat{w}, \mathcal{I})$
5: $g_w \leftarrow g_w^l + \alpha g_w^{reg}$
6: $w \leftarrow w - \eta \cdot \text{Adam}(w, g_w)$
7: **end while**

a straightforward rule is followed for parameter choice: parameters of one layer are chosen if the layer is followed by a randomly initialized layer. In this article, parameters of the 36th, 61th, and 74th (convolutional) layers are chosen when network is pretrained on IN, while parameters of the 80th, 92th, and 104th (convolutional) layers are chosen when the network is pretrained on COCO.

3) Incremental Computation: To prevent underflow or overflow, FIM $\mathcal{I}$ is computed in an incremental style. Specifically, denote the ith data point as $z_i = (I_S^{(i)}, B_S^{(i)}) \in D_S$, and the corresponding loss function l_S for pretraining, $\mathcal{I}$ is iteratively updated as follows:

$$\begin{aligned} g_i &= \nabla l_S(z_i) \\ \mathcal{I}_i &= g_i g_i^T \\ c_i &= 1/i \\ \mathcal{I} &\leftarrow (1 - c_i)\mathcal{I} + c_i \mathcal{I}_i \end{aligned}$$

where $\mathcal{I}$ is initialized as 0.

4) Anchors: As stated in Section II, the sizes of prior anchors is determined by a clustering algorithm. By setting the number of clusters to be 12, we use k-means in scikit-learn package [48] to calculate centroids of sizes of bounding boxes in a training set. For NWPUVHR-10 data set, results are $(18, 18)$, $(31, 20)$, $(22, 30)$, $(36, 34)$, $(44, 45)$, $(64, 32)$, $(61, 65)$, $(106, 46)$, $(94, 119)$, $(155, 75)$, $(110, 164)$, $(157, 136)$. For DOTA data set, results are $(13, 14)$, $(24, 22)$, $(29, 46)$, $(56, 42)$, $(70, 89)$, $(129, 109)$, $(86, 193)$, $(110, 345)$, $(218, 181)$, $(535, 178)$, $(323, 345)$, $(471, 673)$.

5) Matching Strategy: As stated above, it is necessary to determine whether a predicted box is positive, negative, or ignored for each iteration when training. However, there could be ten thousands of predicted boxes for every input image so that the procedure could take lots of time. We find a useful property to optimize the calculation of IoU between different boxes.

Property 3: Given a ground-truth box $B = (x, y, w, h)$ and a predicted box $\hat{B} = (\hat{x}, \hat{y}, \hat{w}, \hat{h})$ of an image, then $\text{IoU}(B, \hat{B}) \leqslant 1/2$ if the predicted center point $(\hat{x}, \hat{y})$ is out of the ground-truth box B.

Proof: As is known, IoU is the rate of intersection area and union area of two boxes. The greater the intersection area is, the greater the IoU is; and the less the union area is, the greater the IoU is. Thus, IoU with the predicted center $(\hat{x}, \hat{y})$ outside B would not be greater than IoU with $(\hat{x}, \hat{y})$ lying at the edge of B (see Fig. 4). Without loss of generality, it is assumed that the predicted center is at the right edge of B. Under the assumption, mathematically, IoU could be expressed as

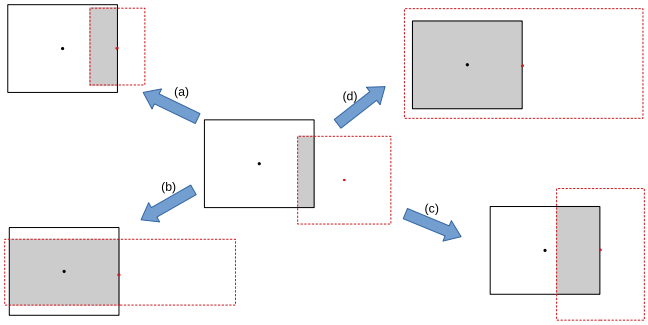


Fig. 4. Different cases of two boxes. The black box is the ground-truth box and the red box with dashed line is the predicted one. There are only four cases under the conditions of Property 3. (a) $\hat{w} < 2w, \hat{h} < h$. (b) $\hat{w} \geqslant 2w, \hat{h} < h$. (c) $\hat{w} < 2w, \hat{h} \geqslant h$. (d) $\hat{w} \geqslant 2w, \hat{h} \geqslant h$.

$$\begin{aligned} &\text{IoU}\left(B, \hat{B}\right) \\ &= \frac{\min\left(w, \frac{1}{2}\hat{w}\right) \min\left(h, \hat{h}\right)}{wh + \hat{w}\hat{h} - \min\left(w, \frac{1}{2}\hat{w}\right) \min\left(h, \hat{h}\right)} \\ &= \frac{1}{\frac{wh}{w \min\left(1, \frac{\hat{w}}{2w}\right) h \min\left(1, \frac{\hat{h}}{h}\right)} + \frac{\hat{w}\hat{h}}{\frac{1}{2}\hat{w} \min\left(\frac{2w}{\hat{w}}, 1\right) \hat{h} \min\left(\frac{h}{\hat{h}}, 1\right)} - 1} \\ &= \frac{1}{\frac{1}{\min\left(1, \frac{\hat{w}}{2w}\right) \min\left(1, \frac{\hat{h}}{h}\right)} + \frac{2}{\min\left(\frac{2w}{\hat{w}}, 1\right) \min\left(\frac{h}{\hat{h}}, 1\right)} - 1}. \end{aligned} \tag{13}$$

Then let $a = \hat{w}/2w$ and $b = \hat{h}/h$, (13) could be written as

$$\text{IoU}(a, b) = \frac{1}{\frac{1}{\min(1, a) \min(1, b)} + \frac{2}{\min\left(\frac{1}{a}, 1\right) \min\left(\frac{1}{b}, 1\right)} - 1}. \tag{14}$$

Consequently, there are four different cases: (a) $a < 1, b < 1$; (b) $a \geqslant 1, b < 1$; (c) $a < 1, b \geqslant 1$; (d) $a \geqslant 1, b \geqslant 1$. For case (a), from (14), it could be derived that $\text{IoU}(a, b) = (1/(1/ab + 1)) \leqslant 1/2$.

The same procedure could be adapted easily to obtain the result for the other cases. ■

Based on Property 3, we set the ignored threshold to be 0.5 and then it only needs to calculate IoU for hundreds of or even tens of times to distinguish predicted boxes when training. For a given ground truth box, matching time with (w/) and without (w/o) the trick when training is shown in Fig. 5. It could be seen that matching w/ the trick only takes up 0.377 ms, while matching w/o the trick takes up 3.702 ms, which is almost ten times as great as the former one.

6) Post-Processing: It is essential to perform nonmaximum suppression (NMS) [49], [50] during inference due to the large amount of generated bounding boxes. In this article, bounding boxes with confidence lower than 0.3 would be filtered out first because they are more likely to be background. Then boxes would be filtered further by NMS with IoU threshold 0.4. The remaining boxes are treated as predictions finally.

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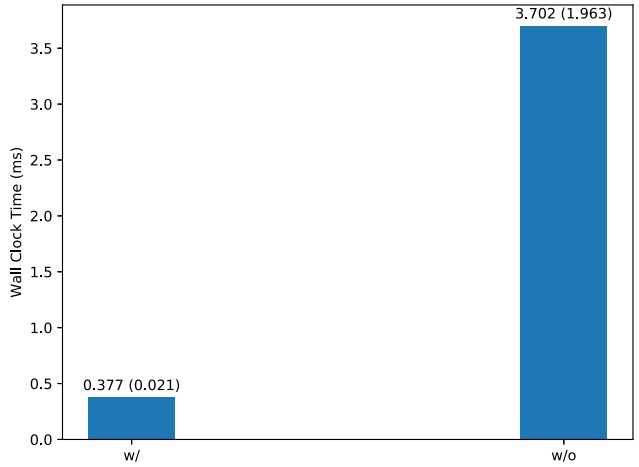


Fig. 5. Time comparison of matching strategies w/ and w/o the proposed trick. Mean and standard deviation (in parentheses) are computed based on 15 randomly picked ground-truth boxes.

7) Evaluation: To evaluate different methods, we adopt a widely used evaluation indicator, mean average precision (mAP), the mean value of AP over all object categories. Specifically, predicted bounding boxes are first divided into four groups: true positive (TP), false positive (FP), true negative (TN), and false negative (FN) according to an IoU threshold (0.5 in this article). Then they are used to calculate precision and recall

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$
$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}.$$

And AP is then obtained by interpolating the precision at each recall level r_{n+1}:

$$\text{AP} = \sum_{n} (r_{n+1} - r_n) \rho_{\text{interp}}(r_{n+1})$$

with

$$\rho_{\text{interp}}(r_{n+1}) = \max_{\tilde{r}:\tilde{r} \geqslant r_{n+1}} \rho(\tilde{r})$$

where $\rho(r)$ is the measured precision at recall r. From the equation, it could be seen that AP is an estimated area under the precision-recall (PR) curve. The PR curve would be used for error diagnosis in Section IV-C.

IV. Experiments

In this section, the effectiveness of the proposed FisherReg would be evaluated from different aspects. First, comparison among different methods is given. Second, error diagnosis is conducted to analyze the proposed method in detail. Third, sensitivity to the choice of hyperparameters is analyzed. Finally, ablation study is conducted to analyze the necessity of different components.

A. Experiment Settings

1) Data Sets: Two public data sets in the literature are used: a small but classical data set, NWPUVHR-10 [18], [39] and a large and challenging data set, DOTA [17]. Statistics and samples of these two data sets are shown in Table I and Fig. 6, respectively.

TABLE I
Statistics of NWPUVHR-10 and DOTA Data Sets.

Dataset	#Categories	#Instances	#Images	Image width
NWPUVHR-10	10	3,651	800	$\sim 1,000$
DOTA	15	188,282	2,806	$800 \sim 4,000$

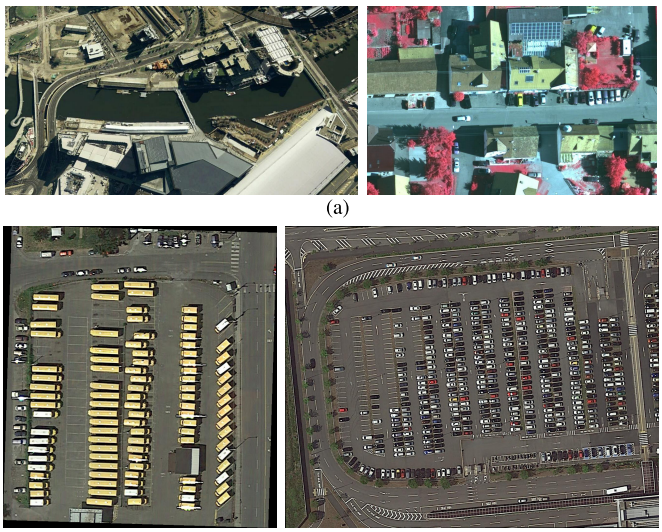


Fig. 6. Samples from NWPUVHR-10 and DOTA data set. (a) NWPUVHR-10. Left: bridge; right: (small) vehicle. (b) DOTA. Left: large vehicle; right: small vehicle.

1) NWPUVHR-10 data set is a publicly available geospatial data set. This data set contains totally 800 optical RSIs in very high resolution and 10 common object categories totally, from which 757 airplanes, 302 ships, 655 storage tanks, 390 baseball diamonds, 524 tennis courts, 159 basketball courts, 163 ground track fields, 224 harbors, 124 bridges, and 477 (small) vehicles are manually annotated with horizontal bounding boxes (HBBs) as ground truth. Usually, these images are randomly split into three parts, and one is training set with proportion of 60%, another is validation set with proportion of 20%, and the last one is test set with proportion of 20%.
2) DOTA data set is a large-scale data set for object detection in aerial images. It contains 2806 aerial images from multiple sensors and platforms and 15 common object categories. The fully annotated DOTA images contains 188 282 instances, each of which is labeled by an arbitrary quadrilateral. Each image is of the size in the range from about 800×800 to 4000×4000 pixels and contains objects exhibiting a wide variety of scales, orientations, and shapes. Half of these images are randomly selected as training set, $1/6$ as validation set and $1/3$ as test set. DOTA is used for two types of object detection, with one for HBB and the other for

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TABLE II
EVALUATION RESULTS ON NWPUVHR-10 DATA SET (%)

Methods		mAP	PL	SH	ST	BD	TC	BC	GTF	HA	BR	SV
Init.	Reg.											
IN	—	90.56	97.85	**94.53**	**97.89**	95.11	89.79	85.29	96.31	93.85	76.46	78.60
	WD	90.08	**99.98**	89.56	96.88	95.22	**90.19**	85.53	95.34	90.80	76.20	81.11
	FIMK	**92.14**	99.11	92.48	96.98	**96.62**	88.79	84.33	**99.05**	**97.32**	**81.97**	84.71
	FIMD	90.73	**99.98**	89.87	97.09	96.33	86.92	**85.88**	95.72	86.39	79.12	**89.97**
CC	—	91.61	99.10	**94.87**	94.43	92.91	**94.21**	87.95	96.59	86.50	80.01	89.57
	WD	91.65	97.55	90.62	95.03	93.81	91.49	80.40	**99.77**	88.47	**90.55**	88.81
	FIMK	92.09	**99.99**	93.50	96.89	93.54	93.23	**88.24**	96.43	**92.57**	72.72	**93.75**
	FIMD	**92.51**	99.95	92.69	**97.81**	**95.53**	91.96	87.16	**99.77**	92.38	79.48	88.35
Faster R-CNN		80.90	94.60	82.30	65.32	95.50	81.90	89.70	92.40	72.40	57.50	77.80
Deformable Faster R-CNN		84.40	90.70	87.10	70.50	89.50	89.30	87.30	97.20	73.50	69.90	88.80
RDAS512		89.50	99.60	85.50	89.00	95.00	89.60	**94.80**	95.30	82.60	**77.20**	86.50
Multi-Scale CNN		89.60	99.30	**92.00**	83.20	**97.20**	**90.80**	92.60	98.10	85.10	71.90	85.90
R^2CNN++		**91.75**	**100.00**	89.41	**97.22**	97.00	83.15	87.54	**99.17**	**99.40**	74.51	**90.10**

oriented bounding boxes (OBBs). In this article, we pay particular attention to the HBB type.

2) Compared Methods: Different methods for object detection are compared on NWPUVHR-10 and DOTA, respectively. On NWPUVHR-10 data set, the proposed method is compared with other five methods, Faster R-CNN [5], Deformable Faster R-CNN [51], RDAS512 [52], Multiscale CNN [53], and R^2CNN++ [10]. On DOTA data set, the proposed method is compared with other seven methods, SSD [11], YOLOv2 [7], region-based fully convolutional networks (R-FCN) [51], faster region-based convolutional neural networks by He (FR-H) [5], FPN [15], image cascade networks (ICN) [54], and R^2CNN++ [10].

3) Hyperparameters: In training stage, different initialization methods are leveraged: random initialization (R) method (weights sampled from a Gaussian distribution, $\mathcal{N}(0, 0.02)$); IN initialization method (weights pretrained on IN data set); COCO initialization (CC) (weights pretrained on COCO data set). Limited by the amount of GPU memory, the number of images of a mini-batch is assigned to be 12. Adam [55] is chosen as the training optimizer and learning rate is assigned to be 1×10^{-4} and changes to be 1×10^{-5} for the last several epochs. Hyperparameters for regularization vary in a range and the best ones on validation set are chosen. Hyperparameters sensitivity is shown in Section IV-D.

All experiments are implemented on the deep learning framework, PyTorch [56]. All codes run on an Nvidia GTX 1080Ti GPU and an Intel Core i7 6700 CPU. Codes would be released soon at our GitHub homepage.[1]

B. Baseline and Comparison

The baseline model is YOLOv3 with modifications described in Section II-A, and it is trained without any regularization. Then it is retrained with the proposed FisherReg to see whether the regularization could improve detection performance or not. The baseline model is also trained with a classical regularization method, weight decay, for comparison.

[1]https://github.com/wasaCheney/FisherReg

All of comparisons are shown in Tables[2] II and III, from which conclusions are drawn as follows. It should be noticed that it might take months to train and test all compared methods so that we simply cite results of compared methods.

It is very hard to train networks from scratch. When networks are initialized with a Gaussian distribution $\mathcal{N}(0, 0.02)$, both mAPs on NWPUVHR-10 and DOTA data sets trivially equal 0. That is to say, the trained networks could detect nothing correctly and therefore the corresponding results are not recorded in tables. It may be caused by the insufficient training data or the insufficient training steps.

Without any regularization, the baseline model with pretraining initialization could achieve very high mAP. On NWPUVHR-10 data set, the network with IN and COCO initialization could get mAP of 90.56% and 91.61%, respectively. It performs better than all compared methods except R^2CNN++. Similarly, on DOTA data set, the network could get mAP of 62.98% and 64.77%, respectively, which performs better than compared one-stage methods (SSD and YOLOv2) with significant margins. Compared with random initialization, in which mAP is 0 in our experiments, these results imply that underlying knowledge from pretraining could be leveraged by simple weight-copying strategy.

Compared with weight decay, the proposed FisherReg could get higher mAP. On NWPUVHR-10 data set, under IN initialization, the proposed FisherReg, performs better than weight decay and the highest mAP is 92.14%, with FIMK as regularization; under COCO initialization, situation is similar and the highest mAP is 92.51%, with FIMD as regularization. Similarly, on DOTA data set, the highest mAPs under IN and COCO initialization are 64.24% and 66.35%, respectively, and both are achieved by FIMK regularization. This result suggests

[2]Abbreviations: Init.—Initialization, Reg.—Regularization, R—Random, IN—ImageNet, CC—COCO, DT—DOTA, WD—weight decay, FIMK—FIM-keynode, FIMD—FIM-diagonal, PL—Plane, SH—Ship, ST—Storage Tank, BD—Baseball Diamond, TC—Tennis Court, BC—Basketball Court, GTF—Ground Track Field, HA—Harbor, BR—Bridge, SV—Small Vehicle, LV—Large Vehicle, RA—Roundabout, SP—Swimming Pool, HC—Helicopter, SBF — Soccer Ball Field. The same below.

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TABLE III
EVALUATION RESULTS ON DOTA DATA SET (%)

Methods		mAP	PL	SH	ST	BD	TC	BC	GTF	HA	BR	SV	LV	RA	SP	HC	SBF
Init.	Reg.																
IN	—	62.98	86.22	83.97	57.85	71.71	**93.82**	64.61	55.21	69.20	37.28	63.23	71.73	58.43	**42.39**	40.08	49.04
	WD	63.65	**87.11**	**85.05**	58.39	69.71	92.65	66.65	53.50	**71.06**	34.64	**66.02**	72.03	59.59	40.99	44.03	53.37
	FIMK	**64.24**	87.09	84.89	**59.52**	71.15	93.15	**68.53**	53.59	70.99	**40.61**	65.48	**72.61**	**60.07**	39.71	42.44	53.75
	FIMD	64.01	86.82	84.15	59.19	**72.31**	93.14	64.74	**55.41**	69.33	37.64	65.29	72.13	57.35	40.32	**47.29**	**54.98**
CC	—	64.77	87.17	85.85	58.65	**73.77**	93.38	61.66	58.77	**71.87**	39.77	65.20	71.25	60.52	40.71	43.21	**59.75**
	WD	64.50	87.43	85.03	59.33	72.89	93.42	67.27	57.93	70.52	**40.10**	60.60	72.67	**61.58**	42.29	39.04	57.42
	FIMK	**66.35**	87.35	**86.46**	**59.42**	73.30	**94.25**	**68.23**	**64.37**	71.60	39.99	**65.33**	**73.73**	57.77	44.04	50.37	58.99
	FIMD	65.37	**88.44**	85.47	58.37	73.00	93.86	62.70	60.00	70.19	39.08	64.55	73.37	58.91	**44.51**	**50.50**	57.56
SSD		10.94	44.74	11.34	27.94	11.21	15.59	12.56	6.91	4.55	6.22	2.00	10.24	4.55	0.53	1.01	14.73
YOLOv2		39.20	76.90	52.37	33.91	33.87	61.65	48.54	34.88	36.44	22.73	38.73	32.02	36.44	38.26	11.61	29.27
R-FCN		47.24	79.33	47.29	65.84	44.26	45.66	47.74	53.53	47.23	36.58	39.38	34.15	44.23	50.64	34.90	37.92
FR-H		60.46	80.32	50.04	59.59	77.55	90.41	75.05	68.13	61.69	32.86	53.66	52.49	49.81	56.46	41.85	57.00
FPN		72.00	88.70	**84.50**	82.60	75.10	90.60	81.30	59.20	**76.60**	52.60	69.40	**78.80**	62.10	66.30	60.10	52.50
ICN		72.50	90.00	78.20	84.80	77.70	90.80	79.10	**73.30**	73.50	53.40	**73.50**	65.00	62.10	70.20	58.10	57.20
R^2CNN++		**75.35**	**90.18**	78.06	**86.39**	**81.88**	**90.91**	**82.44**	73.29	75.77	**55.30**	72.09	77.65	**63.45**	**78.21**	**60.11**	**64.53**

that the proposed FisherReg is more efficient than weight decay.

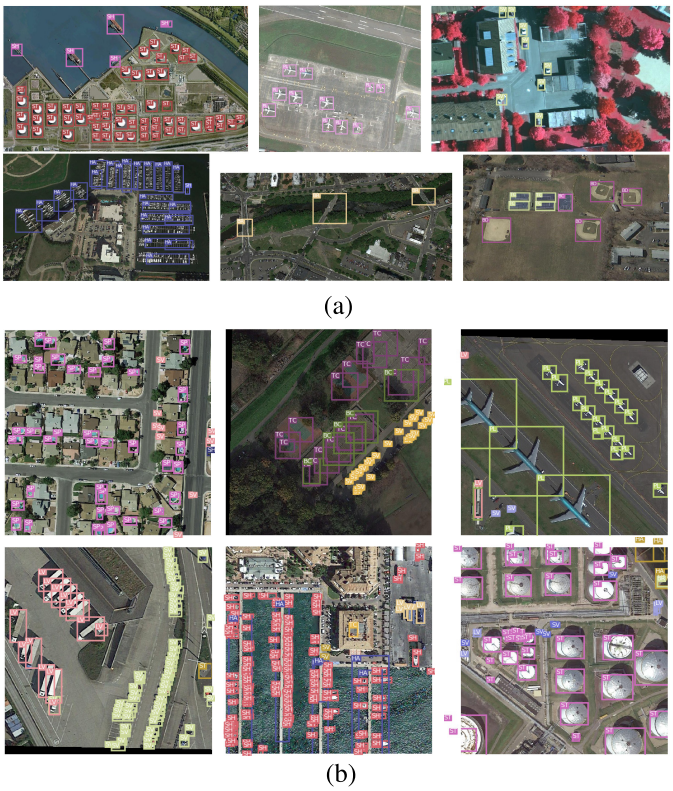
Compared with other methods, the proposed FisherReg could get higher mAP. On NWPUVHR-10 data set, the lowest mAP is 90.08% achieved by the baseline model with IN initialization and weight decay regularization but it is still higher than other five methods except R^2CNN++; the highest mAP is 92.51% achieved by the baseline model with COCO initialization and FIMD regularization and the result is greater than other compared methods. On DOTA data set, the proposed FisherReg is not the best method and its highest mAP is lower than 70%, but it should still be marked as an effective method for several reasons. First, it significantly improves performance of one-stage method, from 10.94% (SSD) and 39.20% (YOLOv2) to 66.35%. Second, it is trained in a single-scale style, while other methods, especially ICN and R2CNN++, are trained with image pyramid. As ablation study shows [10], [54], image pyramid training could offer 5%–7% gain for mAP. In other words, without image pyramids, mAP of ICN and R2CNN++ would decrease to 67%–68%, which is only slightly greater than our method.

C. Error Diagnosis

Fig. 7 illustrates some detection results of the proposed method on NWPUVHR-10 and DOTA data sets, respectively. It shows that the proposed FisherReg is a promising method, especially on NWPUVHR-10 data set, but some objects still could not be detected correctly, especially on DOTA data set. To analyze what mistakes the proposed method tends to make, FPs need to be studied further.

Inspired by [2] and [57], detailed breakdown of FPs of the proposed FisherReg is performed. We reimplement the analysis function in COCO Application Programming Interface (API) but with a little bit modification. As shown in Fig. 8(a) and (b), each plot is corresponding to one category and is composed of a series of PR curves. The meaning of each PR curve is as follows.

1) *AP75*: PR curve at IoU = 0.75.
2) *AP50*: PR curve at IoU = 0.50.



(a)

(b)

Fig. 7. Detection results on different data sets. (a) NWPUVHR-10. (b) DOTA.

3) *Loc*: PR curve at IoU = 0.10. All remaining settings use IoU = 0.10.
4) *Dup*: PR curve after duplicate detections are removed.
5) *BackGround (BG)*: PR curve after all background (and duplicate detections) are removed.
6) *FN*: PR curve after all remaining errors are removed (trivially AP=1).

The corresponding AP is shown in the legend.

Fig. 8(a) shows that on NWPUVHR-10 data set, the proposed FisherReg performs worst on three categories: basketball court, bridge, and vehicle, where AP50 is lower than 0.90. For basketball court, omission (corresponding to FN minus

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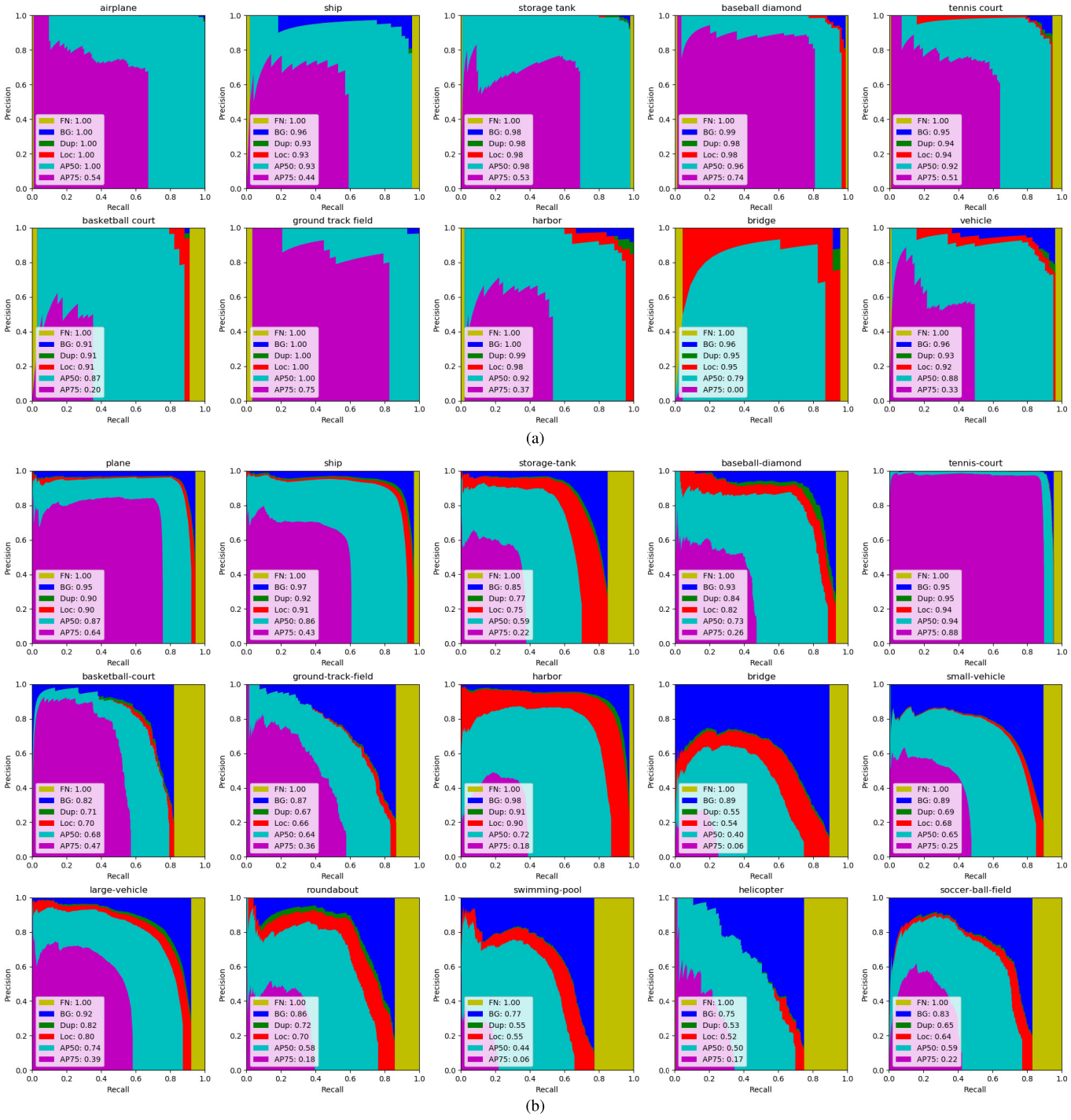


Fig. 8. Error diagnosis on different data sets. See Section IV-C for more details. (a) NWPUVHR-10. (b) DOTA.

BG, the same below) causes decrease of AP by 0.09 and accounts for most part of the error; Location (corresponding to Loc minus AP50, the same below) only causes decrease of AP by 0.04, which is less than half of that caused by omission; However, the proposed FisherReg would not confuse basketball court with background (corresponding to BG minus Dup, the same below) or detect one ground truth repeatedly (corresponding to Dup minus Loc, the same below). For bridge, location causes decrease of AP by 0.16 and it is so influential that any other error could be neglected. For vehicle, omission and location have the same effect on AP and cause decrease by 0.04, respectively; Confusion with background cause slightly lower decrease than omission and location, which is 0.03. In conclusion, the proposed method is more prone to make mistakes of omission and location, but for any category, it would barely make mistake of repeat detection.

Fig. 8(b) shows that on DOTA data set, the proposed FisherReg is prone to make mistakes of confusion with background and omission. For ground track field, bridge, small vehicle, swimming pool, and helicopter, confusion with

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background causes decrease of AP by 0.2, 0.34, 0.2, 0.22, and 0.22, respectively, which is the most dominating mistake. For storage tank, basketball court, swimming pool, helicopter, and soccer ball field, omission causes decrease of AP by 0.15, 0.18, 0.23, 0.25, and 0.17, respectively, which is another noticeable mistake. In other words, the proposed FisherReg cannot clearly tell objects of some specific categories from complicated background, and this phenomenon is worthy of further study.

D. Hyperparameters

In experiments, there is a hyperparameter for regularization term in the objective function. For convenience, hereinafter, λ denotes hyperparameter for weight decay and α for our proposed FisherReg.

To analyze sensitivity of the proposed model about hyperparameters, we conduct experiments with different values of λ and α. Specifically, λ varies in $\{1\times10^{-5}, 1\times10^{-4}, 1\times10^{-3}, 1\times10^{-2}$, and $1\times10^{-1}\}$, and α varies in $\{1\times10^{-2}, 1\times10^{-1}, 1, 1\times10^{1}$, and $1\times10^{2}\}$. Fig. 9 shows the trend of mAP with respect to hyperparameter on NWPUVHR-10 and DOTA data sets.

On NWPUVHR-10 data set, mAP is less sensitive to the proposed FisherReg than weight decay. When $\log\lambda$ increases from -5 to -1, mAP drops rapidly from about 90% to 50%, but whatever $\log\alpha$ is, mAP nearly stays greater than 90%. It could also be seen that under the same regularization, CC initialization performs better than IN initialization, in general. But for an individual initialization method, FIMK and FIMD seem to be similar to each other.

Similarly, on DOTA data set, the proposed FisherReg is still more stable than weight decay. The varied range of mAP for the proposed FisherReg is about from 60% to 67%, while the counterpart of weight decay is about from 27% to 65%. And again, CC initialization outperforms IN initialization for each value of $\log\alpha$.

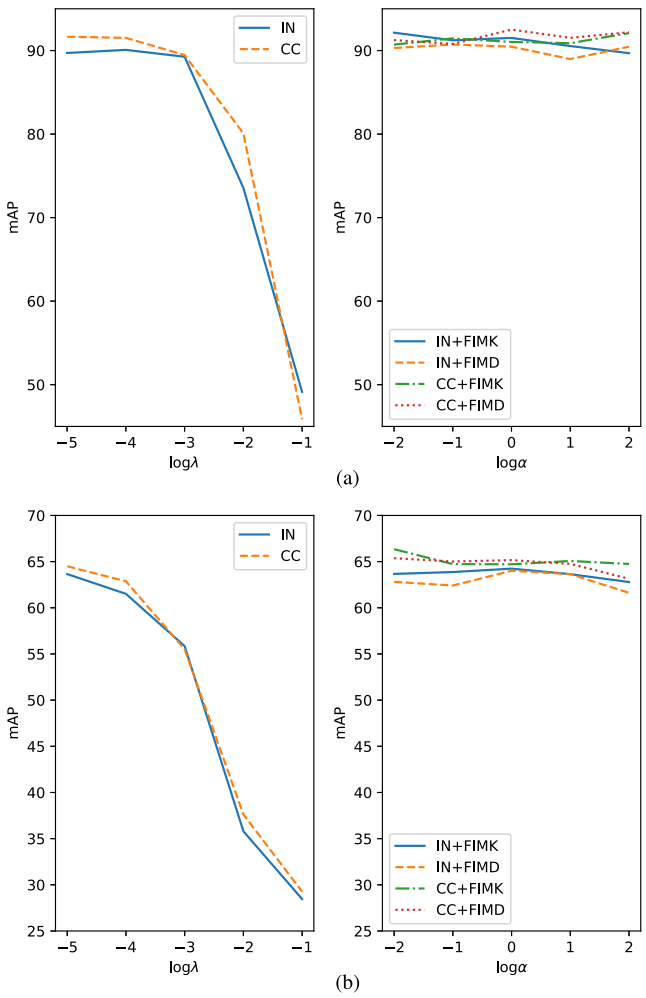


Fig. 9. Sensitivity about hyperparameters on different data sets. In each plot, the left one shows mAP (%) versus λ in logarithmic scale for weight decay, and the right one shows mAP (%) versus α in logarithmic scale for FisherReg. (a) NWPUVHR-10. (b) DOTA.

E. Ablation Study

As stated in Section II-A, the proposed model is based on YOLOv3 with some modifications which include: four anchors in each head instead of three, SmoothL1Loss for width and height prediction instead of MSELoss, and class-balanced loss for imbalanced amount of different kinds of objects.

Ablation experiments on these modifications are conducted to show the corresponding effect. The baseline model is vanilla YOLOv3, and the corresponding component then is substituted one by one gradually.

1) YOLOv3, #Anchors = 3, Loss = MSELoss, w/o balance, and w/o regularization;
2) M1, #Anchors = **4**, Loss = MSELoss, w/o balance, and w/o regularization;
3) M2, #Anchors = 3, Loss = **SmoothL1Loss**, w/o balance, and w/o regularization;
4) M3, #Anchors = 3, Loss = MSELoss, **w/ balance**, and w/o regularization;
5) Ours, #Anchors = **4**, Loss = **SmoothL1Loss**, **w/ balance**, and w/o regularization.

For fairness and simplicity, all resulted models use the same other settings, including IN initialization, Adam optimizer, and so on. Experiments are conducted on NWPUVHR-10 data set. Results are shown in Table IV (left half part). It shows that these modifications could boost performance of the vanilla YOLOv3 apparently. Among these modifications, SmoothL1Loss contributes least, only 0.32% greater than the vanilla YOLOv3, but it is still applied in experiments for it could prevent exploding gradients [4]. The class-balanced loss contributes most, more than 10% greater than the vanilla YOLOv3. That is to say, all three modifications are helpful to improve the detection result.

Furthermore, the proposed FisherReg is applied to other network architectures to verify its effectiveness, that is, the "generalization" to other architectures. We test FIMD style FisherReg on NWPUVHR-10 data set with two popular object detectors: one is one-stage method SSD [11], and another is two-stage method Faster R-CNN [5].

The results are shown in Table IV (right half part). With an SSD detector, the proposed FisherReg improves detection of mAP from 82.78% to 84.66%, and with a Faster R-CNN

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TABLE IV

RESULTS OF ABLATION STUDY ON NWPUVHR-10 DATA SET. LEFT HALF PART SHOWS THE EFFECT OF DIFFERENT COMPONENTS ON YOLOV3 ARCHITECTURE, WHILE THE RIGHT HALF PART SHOWS THE PERFORMANCE OF THE PROPOSED FISHERREG ON OTHERARCHITECTURES. "—" MEANS THE ENTRY IS THE SAME AS THE ONE IN THE COLUMN YOLOV3. SEE MAIN BODY OF THE TEXT FOR MORE DETAILS

Architecture	YOLOv3	M1	M2	M3	Ours	SSD		Faster R-CNN	
#Anchors	3	4	—	—	4				
Loss	MSELoss	—	SmoothL1Loss	—	SmoothL1Loss				
Balance	No	—	—	Yes	Yes				
Regularization	No	—	—	—	—	—	FisherReg	—	FisherReg
mAP (%)	78.25	80.28	78.57	89.17	90.56	82.78	84.66	90.50	91.88

detector, it improves detection of mAP from 90.50% to 91.88%. Both prove that the proposed FisherReg could be applied to different network architectures. It should be noted that here the result of Faster R-CNN is different from that in Table II, for we reimplement the algorithm based on a PyTorch version.[3]

V. CONCLUSION

In the literature of object detection in optical RSIs, it is common to initialize networks by pretrained weights on IN data set for knowledge transferring. However, there are few works analyzing why the paradigm works. He *et al.* [58] compared the paradigm with random initialization and verified its effectiveness experimentally.

In this article, we give an interpretation of the paradigm from the view of Bayesian transfer learning: pretrained weights could be treated as a trivial solution of Fisher regularization term. Moreover, based on FIM, we develop a regularization term, FisherReg, which could make full use of hidden knowledge in pretrained weights.

Experiments on NWPUVHR-10 and DOTA data sets show that the proposed FisherReg is effective and efficient. It could get higher mAP than its counterparts w/ or w/o weight decay and it performs better than compared one-stage methods and goes more closed to two-stage methods. Moreover, FisherReg is parameter-free, less sensitive to hyperparameter than weight decay, and works well with different object detectors.

However, FIM is computationally inefficient so that it is approximated in this article. Though experiments show the approximation works well, but we believe that it is better to find a proper way to compute and store FIM more efficiently.

ACKNOWLEDGMENT

The authors would like to thank the editors and reviewers for their valuable comments and suggestions. The authors would also like to thank Zengjie Song, Rongrong Fei, Kai Sun, and Lizhen Ji for their help and thank the developers and maintainers of PyTorch.

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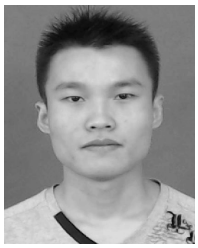
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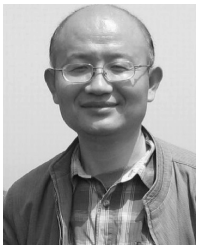
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